Chainalysis

Cryptocurrency Exchanges in 2021

A Competitive Landscape Analysis
November 2021



Table of Contents

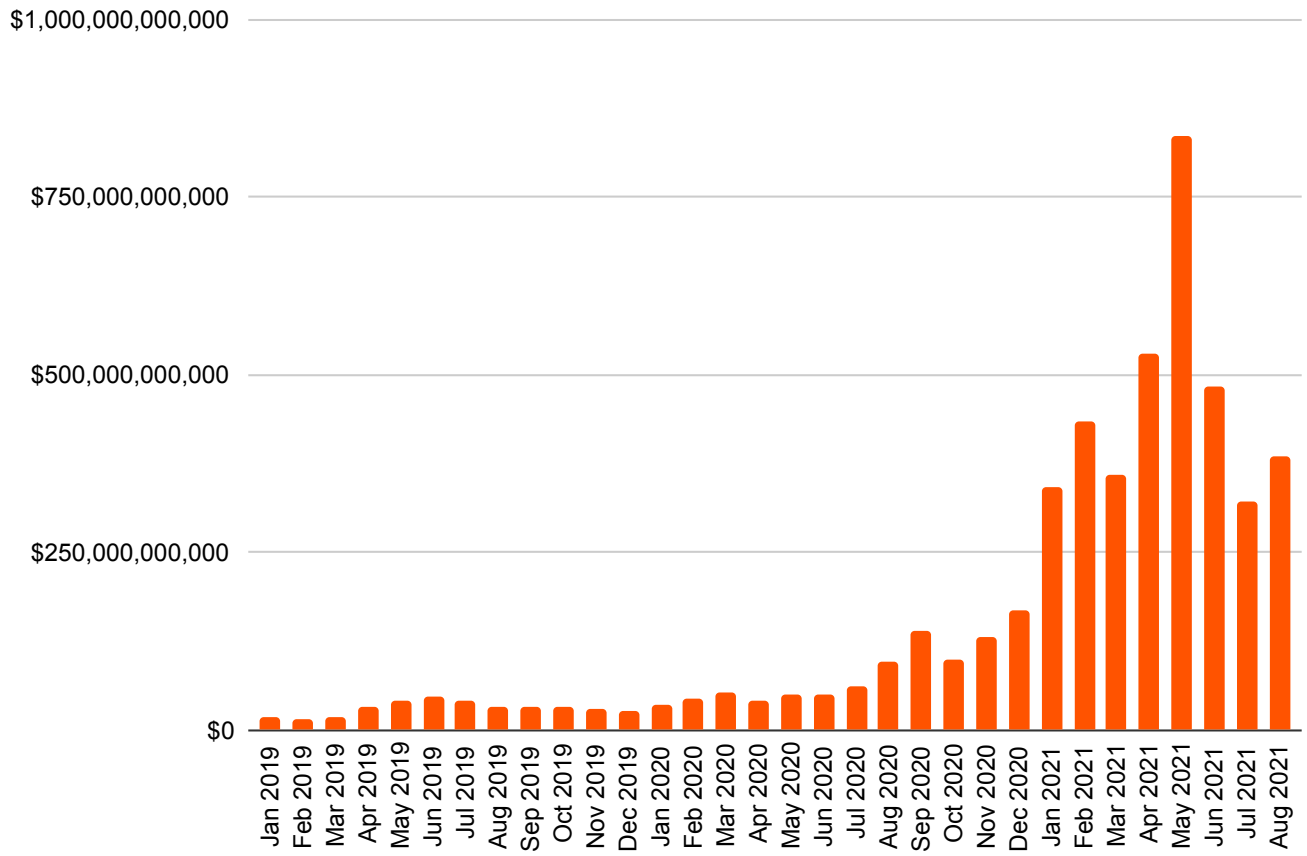
Introduction	3
Breaking exchanges down by business model	4
Diving deeper into exchange size and assets on offer	10
Innovation and scale win in a competitive landscape	13

Introduction

Cryptocurrency adoption and prices have grown steadily over the last few years, allowing the asset class to attract institutional investors and continue gaining steam. Against this backdrop, the businesses making up the cryptocurrency industry have also undergone a period of sustained growth. Check out the graph below, which shows monthly cryptocurrency value received by exchanges since 2019.

Monthly cryptocurrency value received by all exchanges

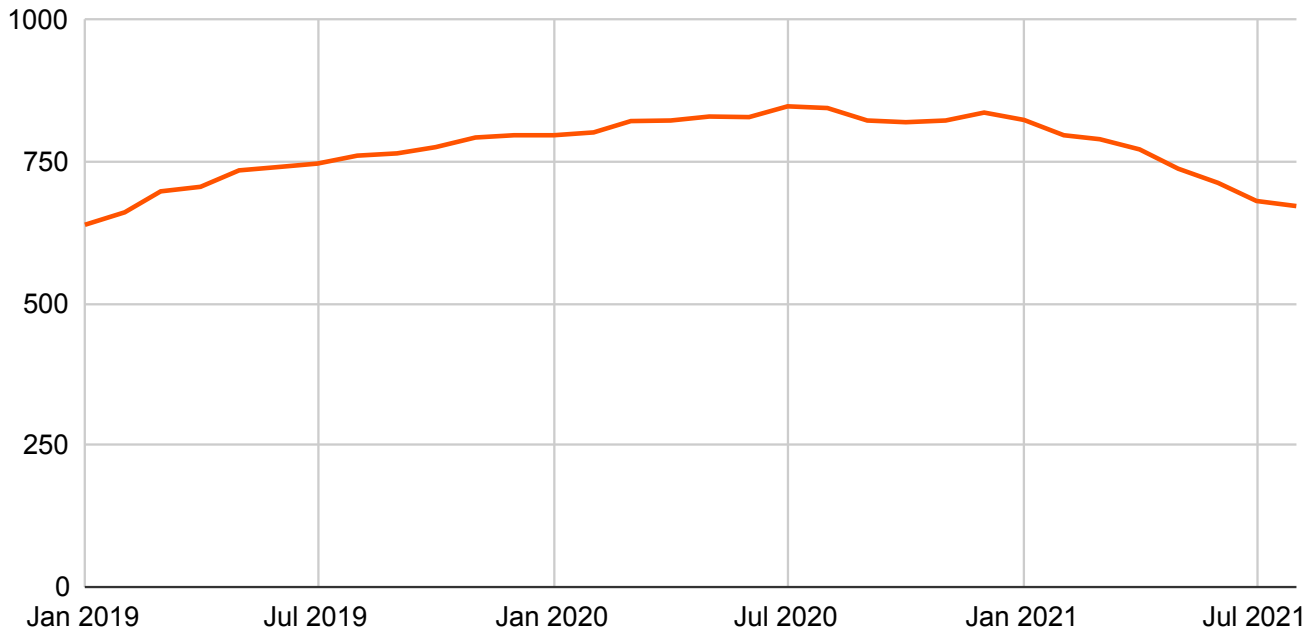
| Jan '19 - Aug '21



While the monthly increases aren't constant, the value moving to exchanges is generally trending upwards. However, over the last year, the cryptocurrency exchange landscape has become extremely competitive and now appears to be consolidating. We can see this on the graph below, which shows the number of active exchanges each month from January 2019 to the present.

Number of active cryptocurrency exchanges by month

| Jan '19 - Jul '21



The number of active exchanges flattens around July 2020 and then starts to fall. It sits at 672 as of August 2021, having peaked at 845 in August 2020. Why is the number of active cryptocurrency exchanges dropping? And what differentiates the most successful of those left standing? In this report, we try to answer those questions by looking at differences in exchange growth trends across a number of variables.

Breaking exchanges down by business model

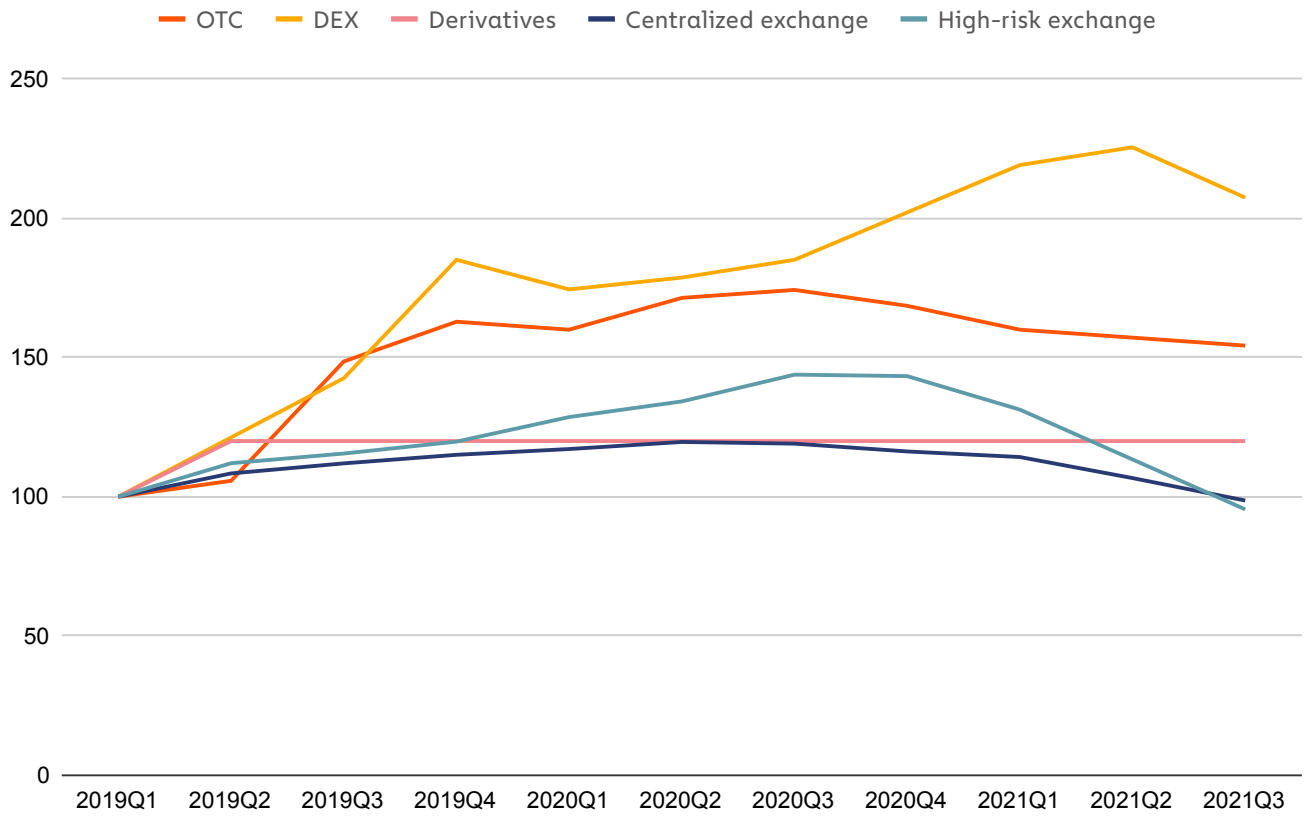
We break exchanges down into six different categories based on their business models and technical infrastructure in order to investigate which types are thriving as the industry becomes more competitive. Those exchange categories are:

- **Centralized exchanges**
- **Decentralized exchanges (DEXes)**
- **High-risk exchanges, meaning those with minimal KYC requirements**
- **OTC brokers**
- **Derivatives exchanges**

Using these categories, let's look at which categories have grown or shrunk by number of active exchanges since the beginning of 2019.

Index: Growth in number of active exchanges by business model

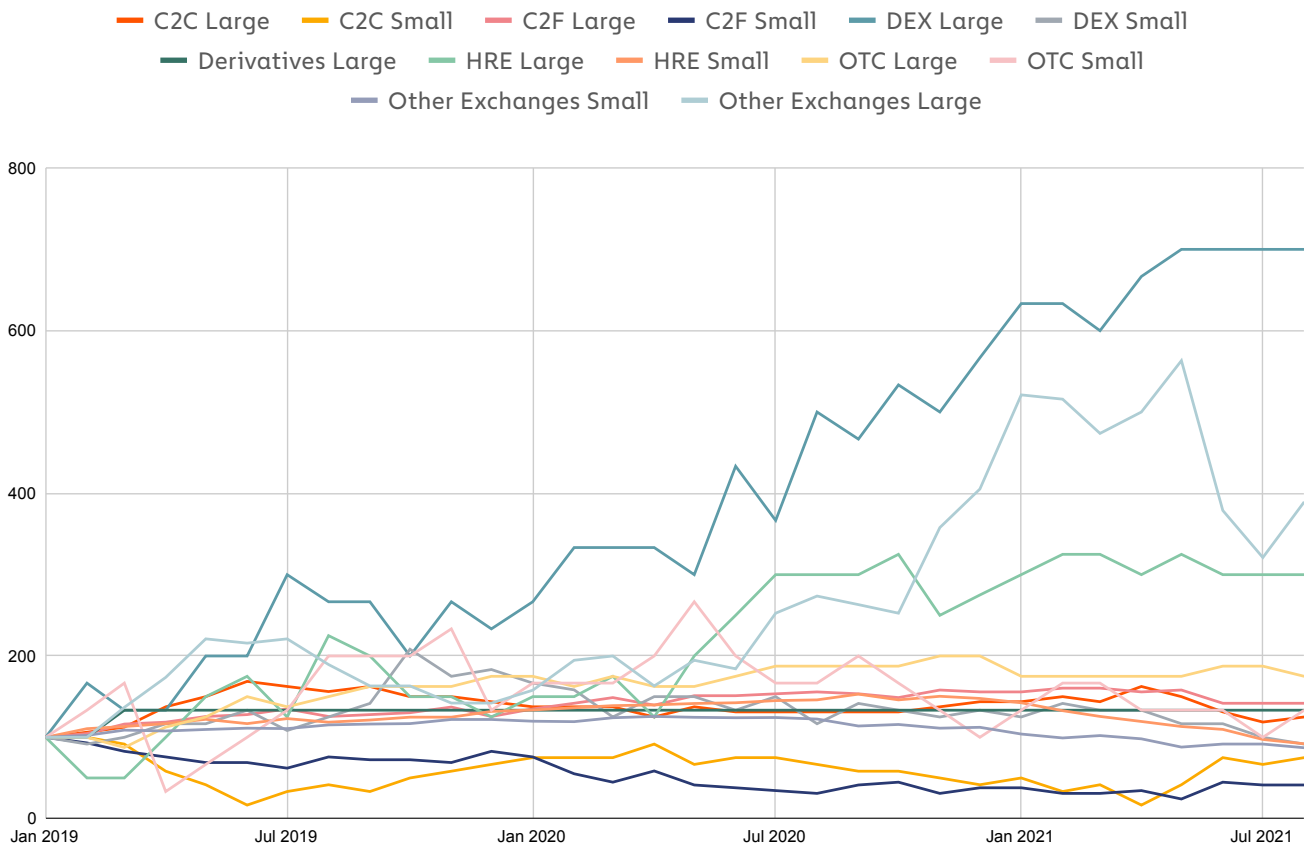
| Apr '19 - Jun '21



The data shows that the number of active DEXes and OTC brokers has climbed significantly since 2019, with derivatives exchanges also growing modestly. Centralized exchanges and high-risk exchanges, meanwhile, have seen their numbers dip slightly after initial increases.

Beyond that, we also further break down each of those categories based on size, characterizing all exchanges as either small or large, with small exchanges being those with under $10 million in cryptocurrency value received between August 2020 and August 2021. Additionally, we can also break out centralized exchanges into more granular categories: Crypto-to-crypto (C2C) exchanges are those that only allow trades between different types of cryptocurrencies, while crypto-to-fiat (C2F) exchanges allow users to exchange fiat currency for cryptocurrency, making them the go-to service for users on or off-ramping into cryptocurrency.

Index: Growth in number of exchanges by business model and exchange size



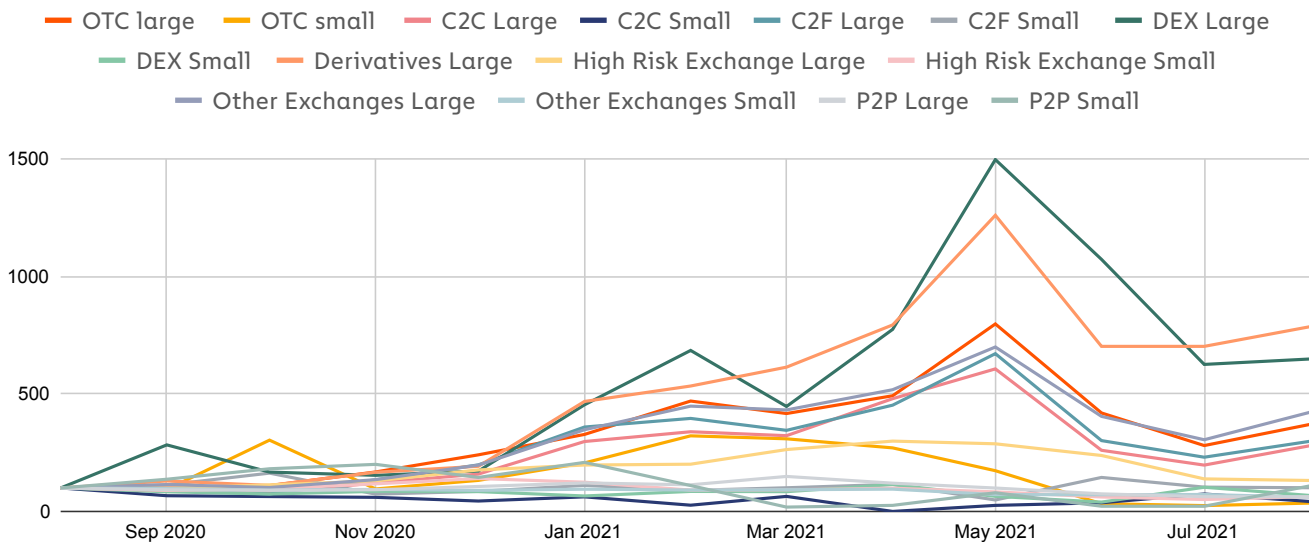
Note: Our analysis includes only four derivatives exchanges all of which qualify as large, hence why that's the only size distinction we make in the category.

Incorporating the size of each type of exchange gives us a more granular view of which kinds of cryptocurrency businesses are thriving. Large DEXes, large high-risk exchanges, and large exchanges in the "other" category — this mostly includes exchanges that are nominally C2F but whose users treat them as C2C — have grown the most by far, with the numbers for all three more than tripling. Several other categories show moderate but substantial growth as well, including large OTCs, large C2F exchanges, and derivatives exchanges. Small exchanges across most business models on the other hand, including C2F and C2C exchanges, saw their numbers shrink.

Of course, the number of active exchanges in each category isn't the only way to judge the health of those categories. After all, cryptocurrency businesses aren't simply trying to survive — they need to grow their userbases and transaction volumes in order to thrive. Below, we look at the same categories by growth in value received.

Growth in received value by market segment

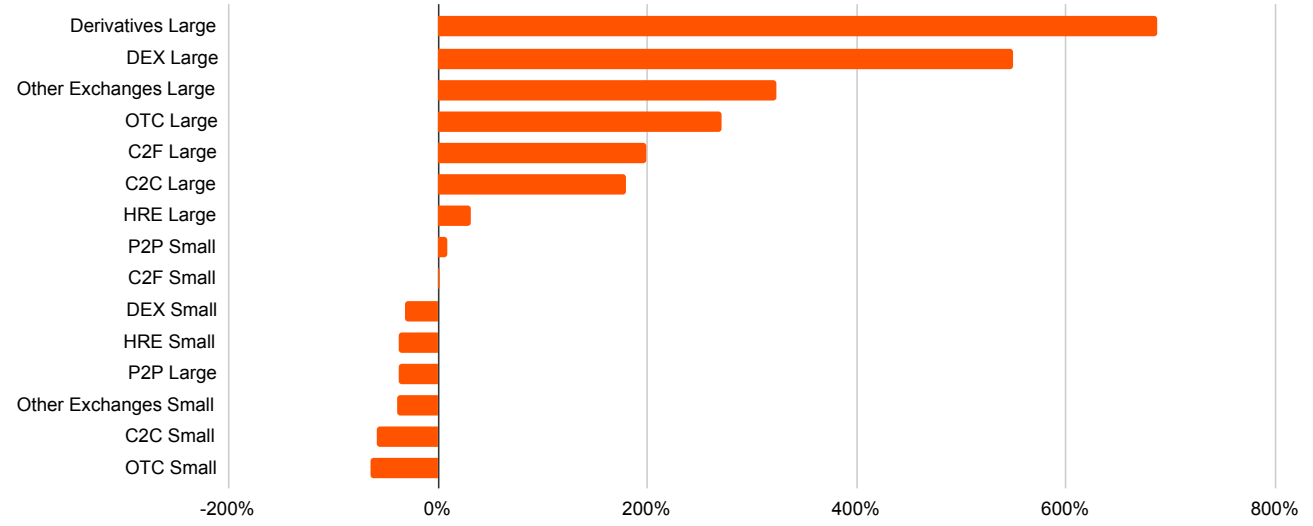
| Apr '19 - Jun '21



One can see the bifurcation between service types that grew substantially and those that shrunk or stayed roughly even. Interestingly, most services saw significant declines in value received between May and June. While we can't say why with total certainty, we believe this may be related to China's decision in May to ban cryptocurrency mining.

Below, we summarize the findings of the chart above by showing each exchange category's total growth in transaction volume between August 2020 and August 2021.

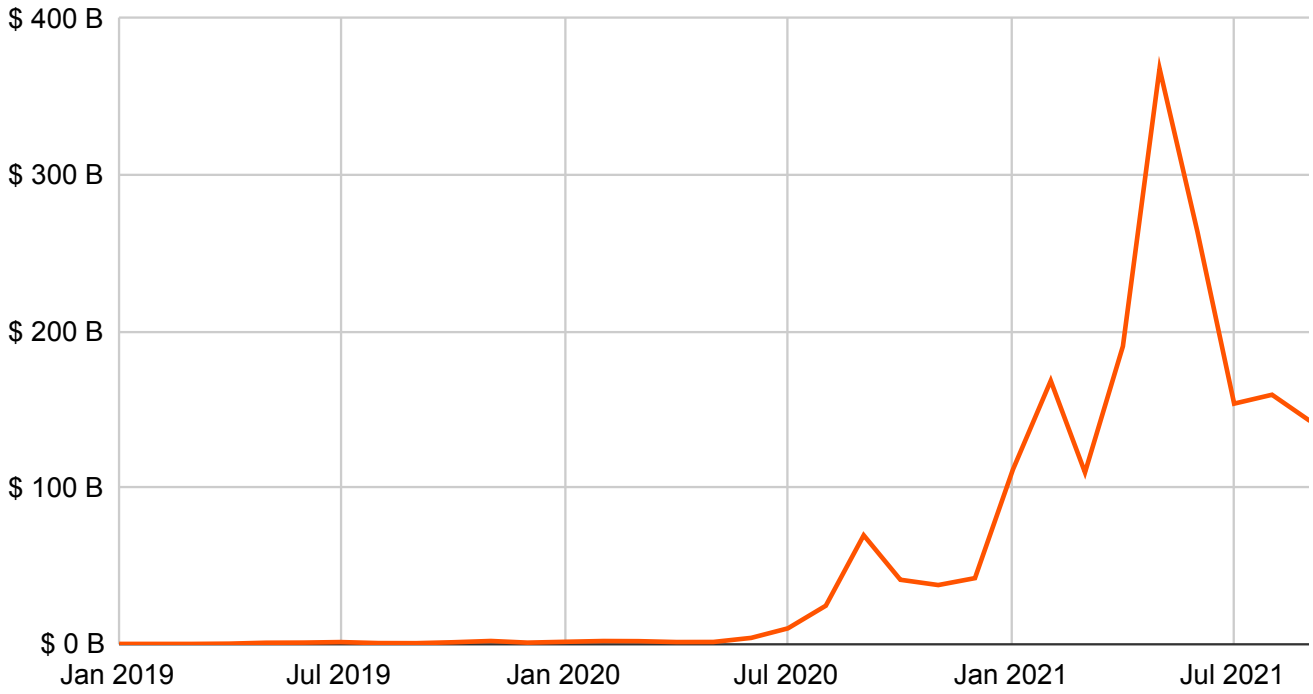
Growth in value received between Aug 2020 and Aug 2021 by market segment



These results confirm some of our previous takeaways. Large DEXes grew their transaction volume substantially during the time period studied, as did large OTC brokers and large centralized exchanges. However, despite not seeing much growth in the number of active businesses, derivatives exchanges grew the most by far in value received at 686%. On the other hand, nearly all small exchange categories regardless of business model saw a decrease in their cryptocurrency received.

The biggest takeaway here is that DEXes have become extremely popular, which coincides with the explosive growth of the DeFi category in general.

Total monthly value received by DEXes



Total value received by DEXes grew from just over $10 billion in July 2020 to a peak of $368 billion in May 2021, and as of September 2021 sat at just under $143 billion. While centralized service transaction volume has also grown, it hasn't grown at nearly the same rate, and during some months DeFi activity has eclipsed activity on centralized services.

In our 2021 Geography of Cryptocurrency report, we discussed the differences between DeFi users and cryptocurrency users generally. Our research shows that DeFi transactions tend to be larger than cryptocurrency transactions on centralized services. Below, we compare DEX transaction sizes to those of a few other exchange types using transactions from August 2021.

August 2021 Transfers	Number of transactions	Mean tx size	25th percentile	50th percentile	75th percentile	Max tx size
OTC broker	60,008	$272,387	$7,373	$15,870	$51,234	**$469,081,200**
DEX	5,152,359	$26,520	$196	$904	$4,200	**$187,675,600**
Derivatives exchange	265,078	$79,190	$130	$489	$3,660	**$708,751,600**
Centralized exchange (all types)	14,669,926	$12,431	$49	$150	$783	**$1,000,500,000**

We see that DEX users carry out much larger transactions than centralized exchange users — the average DEX transaction is over $26,000 worth of cryptocurrency versus over $12,000 for centralized exchanges, while the median for DEXes is just over $900 versus $150 for centralized exchanges. This is likely because DeFi is also more popular in countries with bigger, more established cryptocurrency markets, which also tend to be wealthier countries. Experts we spoke to like David Gogel, Growth Lead of the popular decentralized exchange dydx, explained this dynamic, telling us that most DeFi users are established cryptocurrency investors or traders looking for new sources of alpha. Knowing this, it makes sense that their average transaction sizes would be larger, as DEX users have likely already built up a large base of funds to deploy, or are investing money on others' behalf. The chart above shows that a similar dynamic likely exists with derivatives exchanges, whose average and median transaction sizes are also much larger than those of standard centralized exchanges.

However, despite the growth of DEXes and derivatives exchanges, the more modest growth of large-sized centralized exchanges shows that they still have an important

role to play in the cryptocurrency world. As cryptocurrency adoption grows, most new users will acquire their first cryptocurrency through centralized exchanges, as these are typically the easiest services to exchange fiat currency for different types of cryptocurrency. Furthermore, even experienced traders and DeFi users who want to exchange cryptocurrency for cash will need to rely on these services.

OTC growth could also be a function of more professional and institutional investors moving into the space. OTC brokers frequently help such users execute extremely large trades that would strain the liquidity of open exchanges or possibly affect prices — the growth of OTCs as a category suggests this may be a growing use case.

Diving deeper into exchange size and assets on offer

As the data above suggests, bigger cryptocurrency businesses are growing and surviving at a higher rate than smaller ones. Across all exchange categories, large-sized exchanges were the only group to grow between 2019 and the present, with the sole exception of small P2P platforms, which grew just 9% in value received. This could be for a number of reasons. Perhaps new users are more likely to hear about bigger exchanges, and therefore flock to them when they decide to buy their first cryptocurrency. Or, perhaps bigger exchanges' superior liquidity allows them to attract the biggest traders.

Our analysis suggests that the number of unique crypto assets available plays a big role in exchanges' survival rate during the time period studied. Check out the chart below, which plots exchanges' transaction value received versus number of unique assets available.

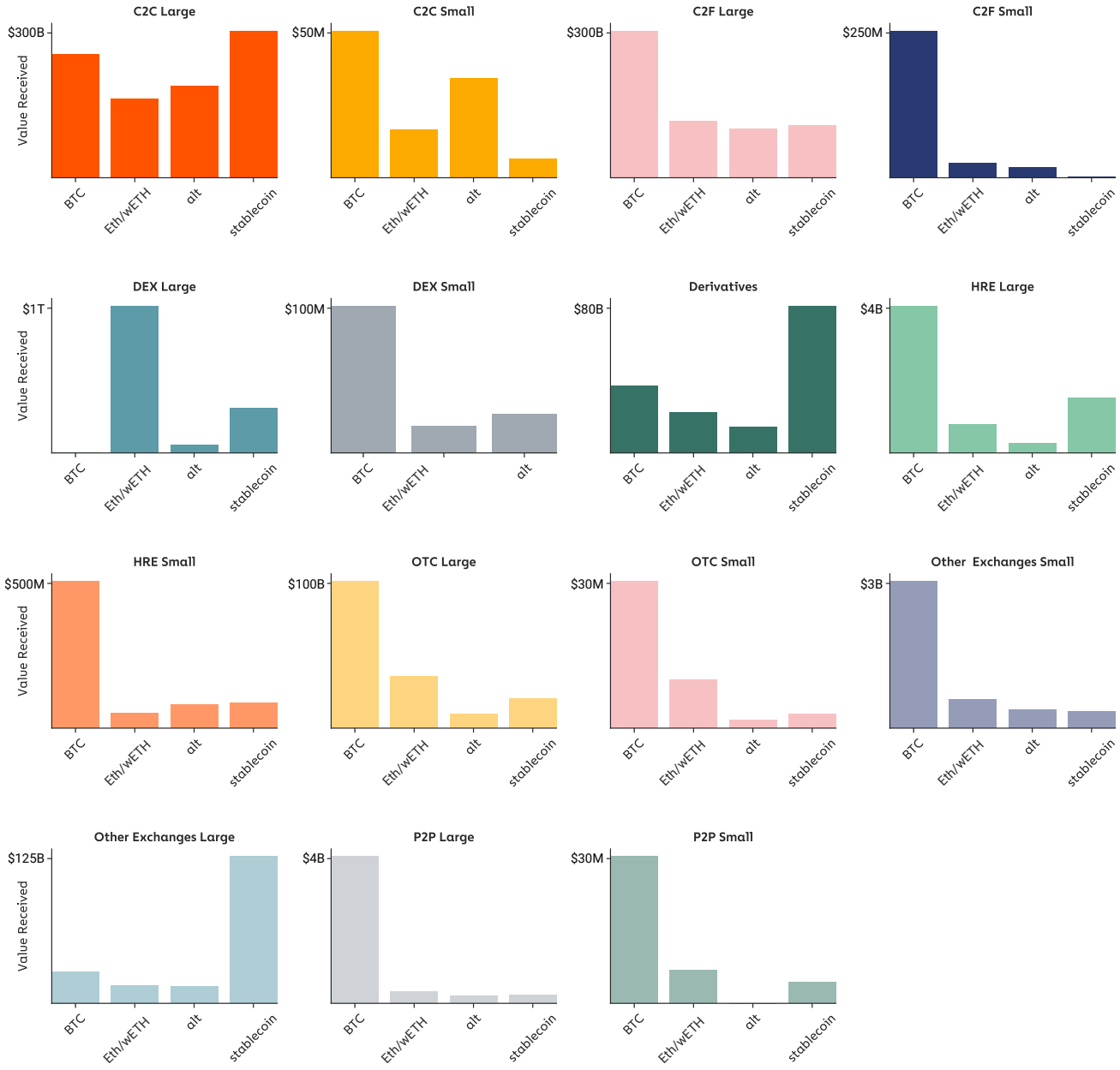
Average monthly value received vs. Number of assets on offer by exchange segment | Aug 2020 - Aug 2021



The data is clear that, while many exchanges with few assets available have high transaction volumes, more unique assets available generally correlates with higher transaction volume.

Below, we break down transaction volume by asset for each exchange category and size.

Total value received by exchange segment and asset class 2021 YTD



Overall, the fastest-growing exchange categories tend to devote most of their transaction volume to Bitcoin or Ethereum. This is relatively unsurprising given that Bitcoin and Ethereum are the two most popular cryptocurrencies for investment.

However, we see some notable exceptions. For instance, derivatives exchanges were the fastest-growing exchange category by value received during the time period studied, and stablecoins are their most-traded asset. The same is true for large exchanges in the "other" category and for large C2C exchanges, which were the third and sixth-fastest

growing categories respectively. The likely reason for this is that both derivatives and C2C exchanges, as well as many in the "other" category, cater to experienced traders and don't allow on-platform conversion into fiat currency. Because of that, traders rely on stablecoins to lock in their assets' value when they no longer want to trade or remain exposed to cryptocurrency price swings.

What stands out most though is that for the most part, the fastest-growing exchange categories tend not to draw all of their transaction volume from any one asset or get very little transaction volume from any one asset. While Bitcoin or Ethereum tend to be the most popular, the fastest-growing exchange categories typically still see significant volume across the other two asset types, suggesting that the best exchanges cater to a variety of use cases, or to use cases requiring multiple asset types.

Innovation and scale win in a competitive landscape

Our analysis of the competitive landscape for cryptocurrency exchanges suggests that innovation and scale — ideally both — are the keys to differentiating and growing in what has recently become a consolidating industry. DEXes represent innovation in cryptocurrency exchanges. The activity of these decentralized, non-custodial platforms was dwarfed by centralized exchanges as recently as last year, but DEXes have since caught up and in some months surpassed centralized service transaction volume by giving users greater control over their assets and access to new types of trades.

The centralized exchanges that continue to grow, on the other hand, appear to be those offering the widest variety of assets, which keeps them attractive to the most active traders. C2F exchanges, however, can be the exception to that trend, as their status as the on and off-ramps between cryptocurrency and fiat mean they'll always be the go-to platform for new users and experienced users who want to convert cryptocurrency into cash.

Now that you know more about the competitive landscape for exchanges, do you want data on how your cryptocurrency business can grow faster? Check out **Chainalysis Business Data**. Chainalysis Business Data uses the same best-in-class dataset behind our investigative and compliance products to help you answer questions like, "Which assets should we run promotions on?" and "Are customers leaving for competitors?" **Contact us here** to set up a demo and see how Chainalysis Business Data can surface the intelligence you need to attract and retain more users.

Chainalysis

ABOUT CHAINALYSIS

Chainalysis is the blockchain analysis company providing data and analysis to government agencies, exchanges, and financial institutions across 40 countries. Our investigation and compliance tools, education, and support create transparency across blockchains so our customers can engage confidently with cryptocurrency. Backed by Accel, Benchmark, and other leading names in venture capital, Chainalysis builds trust in blockchains. For more information, visit www.chainalysis.com.

GET IN TOUCH:

info@chainalysis.com

FOR MORE CONTENT:

visit blog.chainalysis.com

This document is not intended as legal advice. We recommend you consult your general counsel, chief compliance officer, and/or own compliance policies & procedures for regulatory, legal or compliance-related questions.

Building trust in blockchains